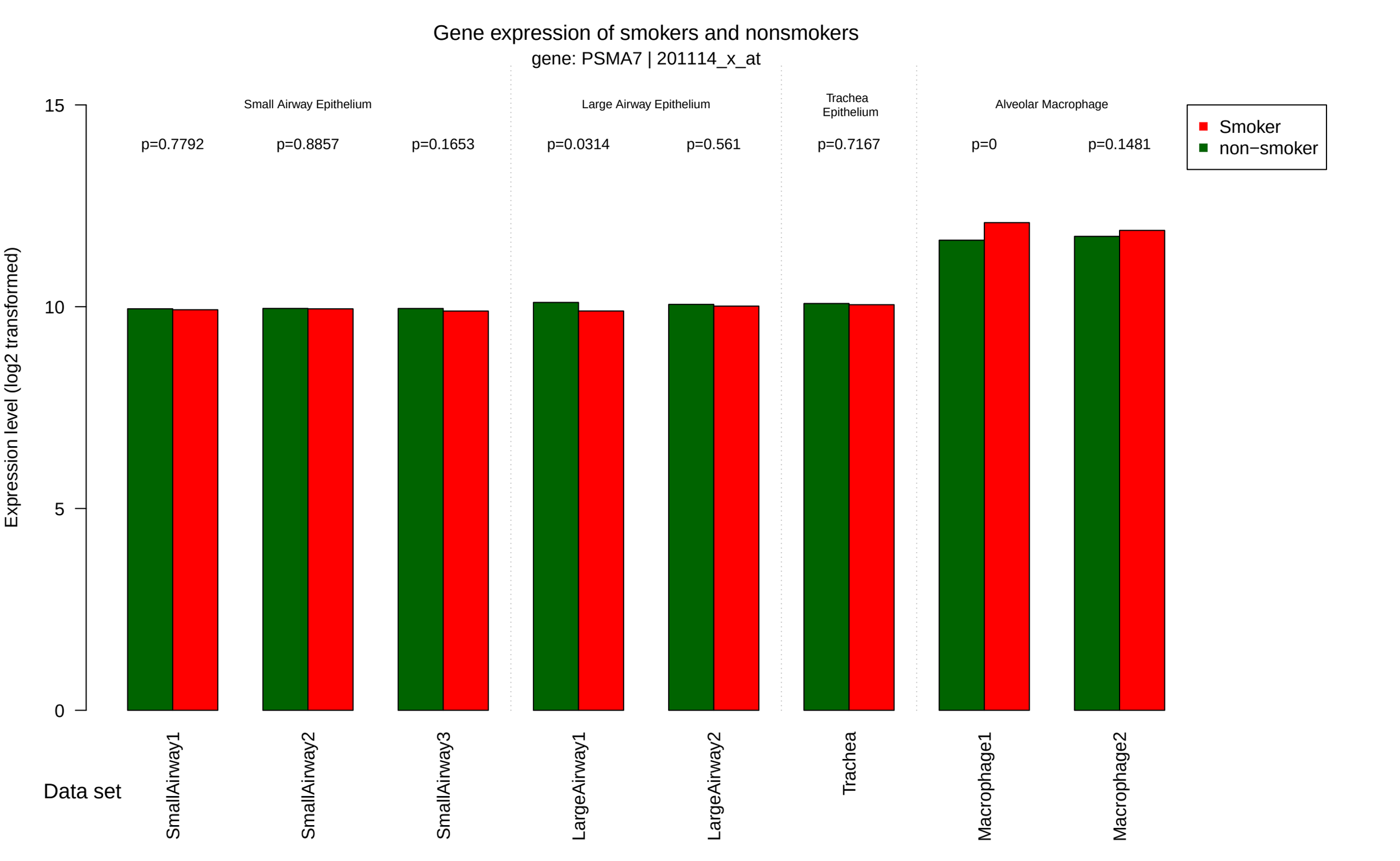
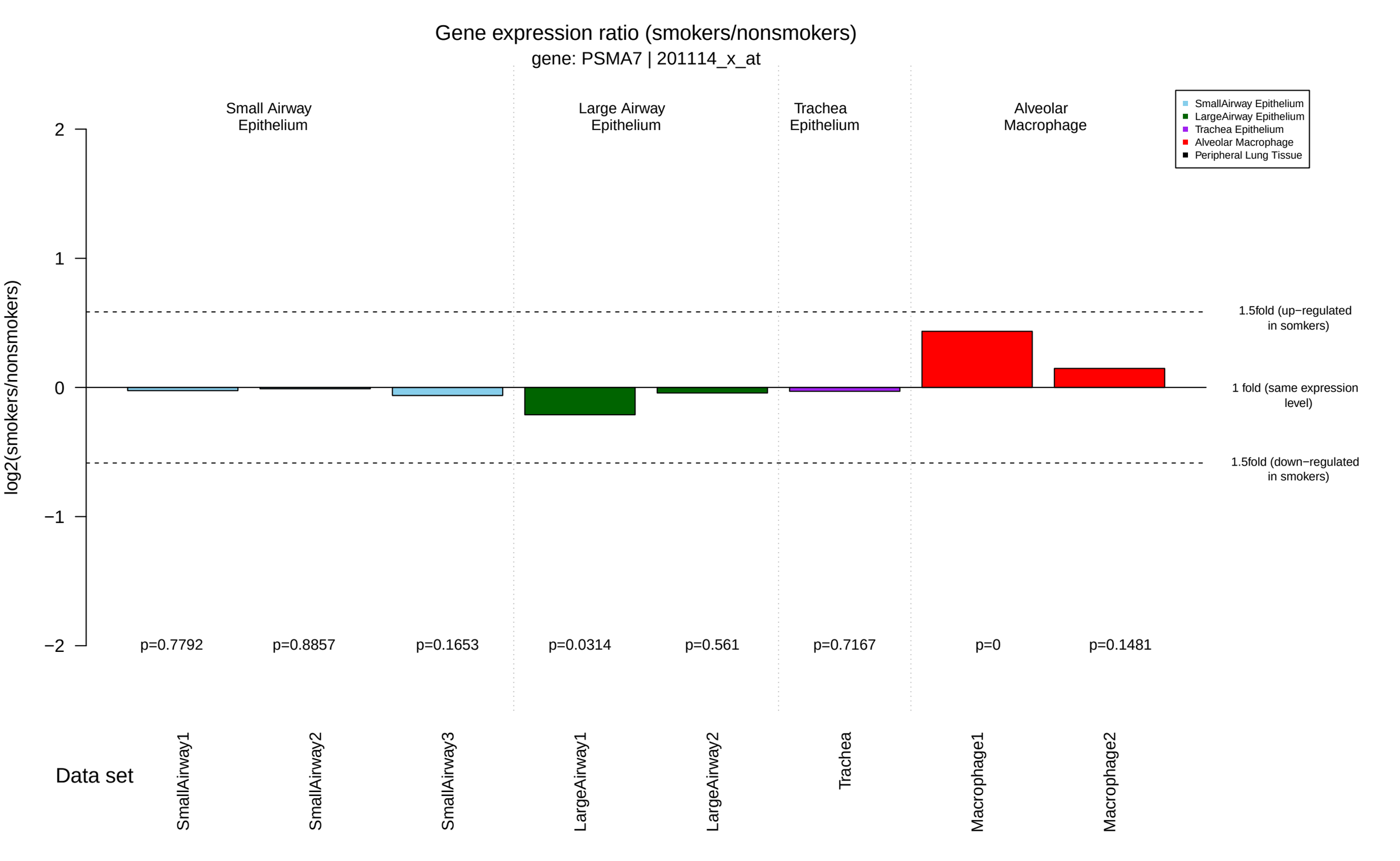
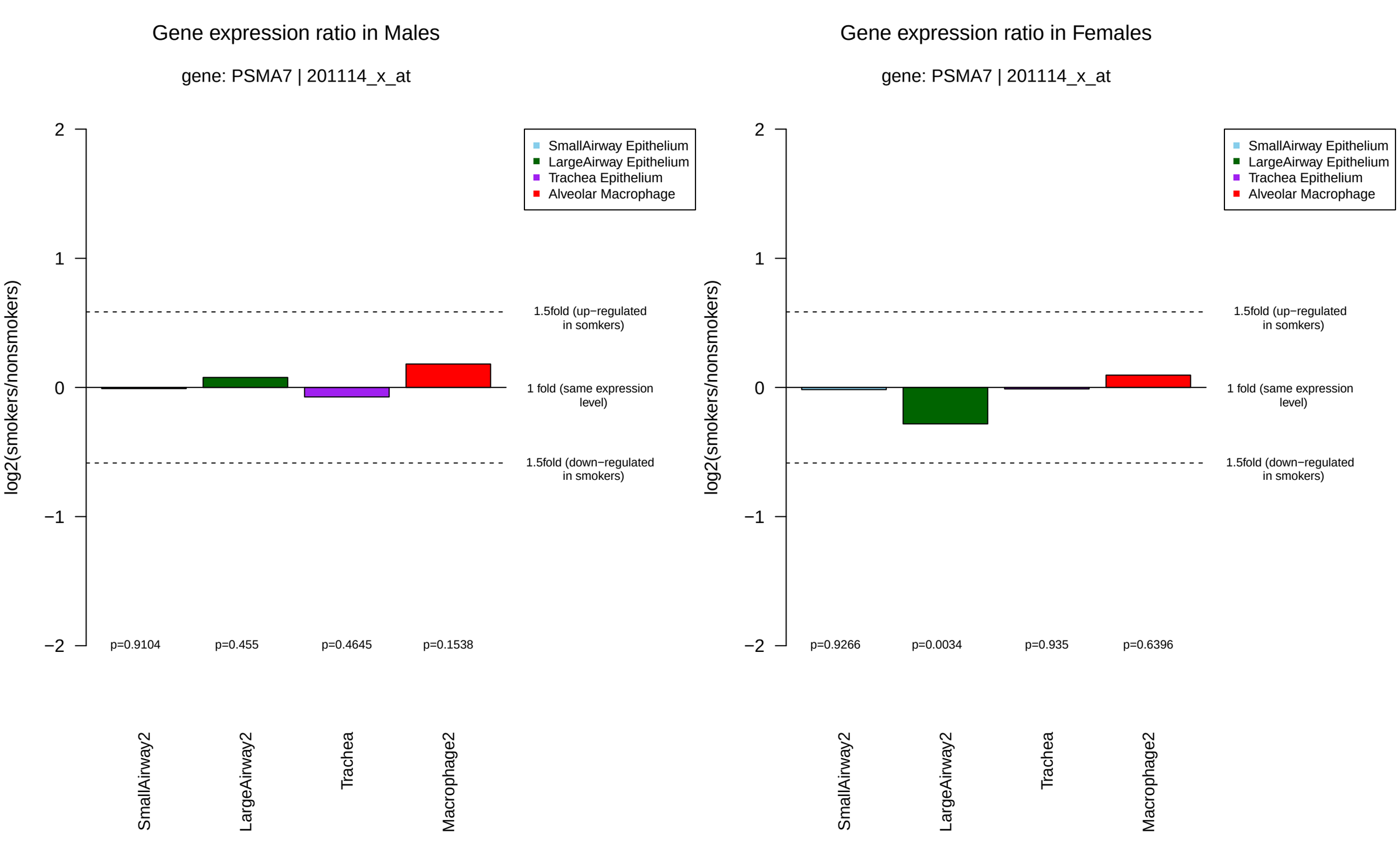
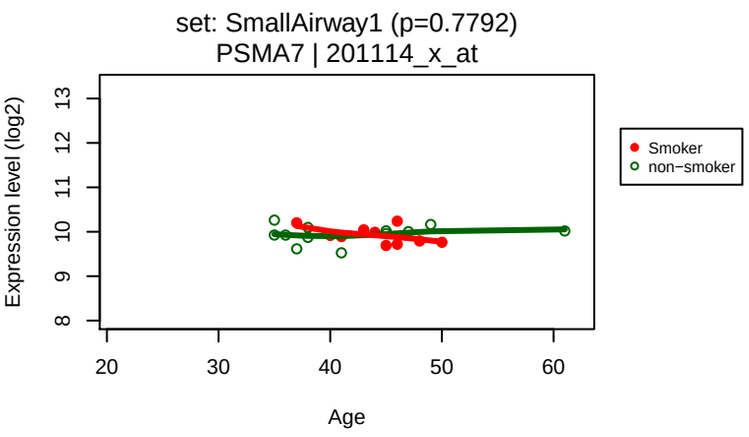
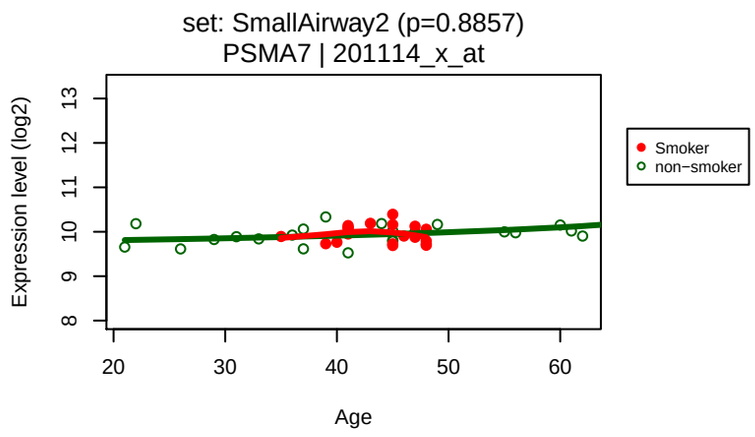
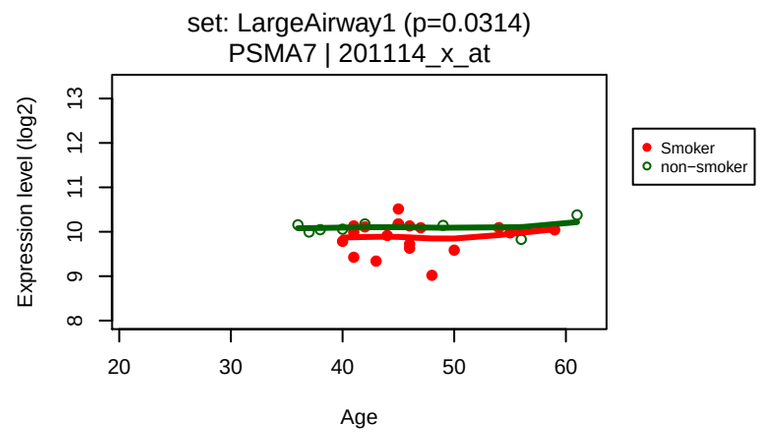

Gene expression of smokers and nonsmokers
gene: PSMA7 | 201114_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7792
p=0.8857
p=0.1653
p=0.0314
p=0.561
p=0.7167
p=0
p=0.1481
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

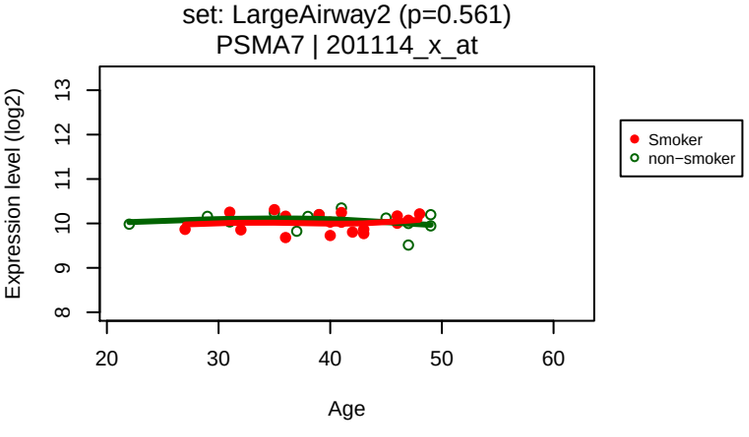

Gene expression ratio (smokers/nonsmokers)
gene: PSMA7 | 201114_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7792
p=0.8857
p=0.1653
p=0.0314
p=0.561
p=0.7167
p=0
p=0.1481
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

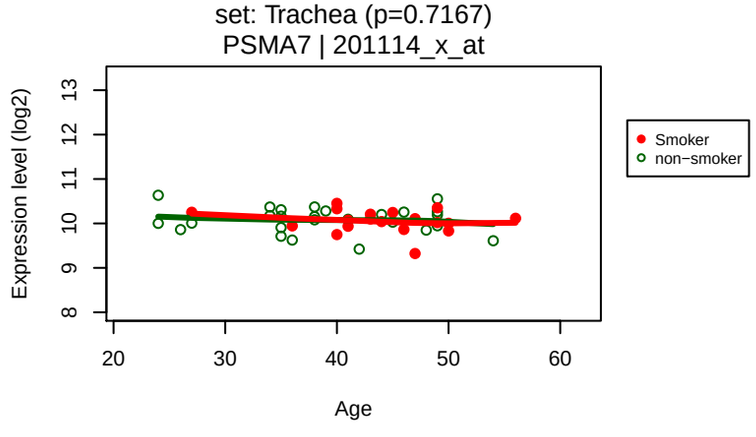

Gene expression ratio in Males
gene: PSMA7 | 201114_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9104
p=0.455
p=0.4645
p=0.1538
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMA7 | 201114_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9266
p=0.0034
p=0.935
p=0.6396
SmallAirway2
LargeAirway2
Trachea
Macrophage2

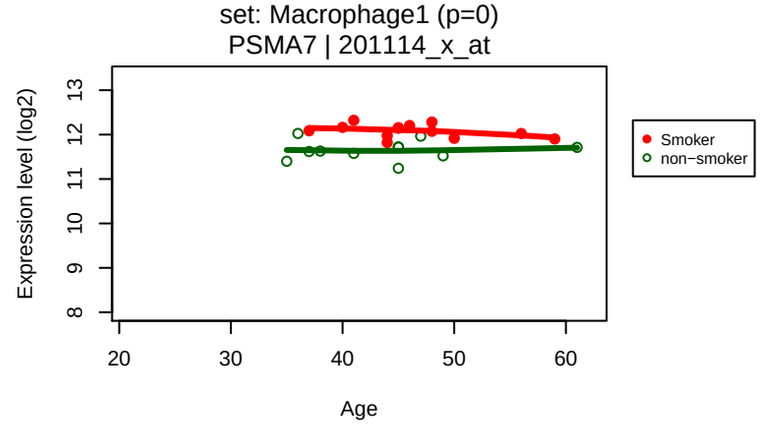

set: SmallAirway1 (p=0.7792)
PSMA7 | 201114_x_at
Expression level (log2)
Age
Smoker
non−smoker

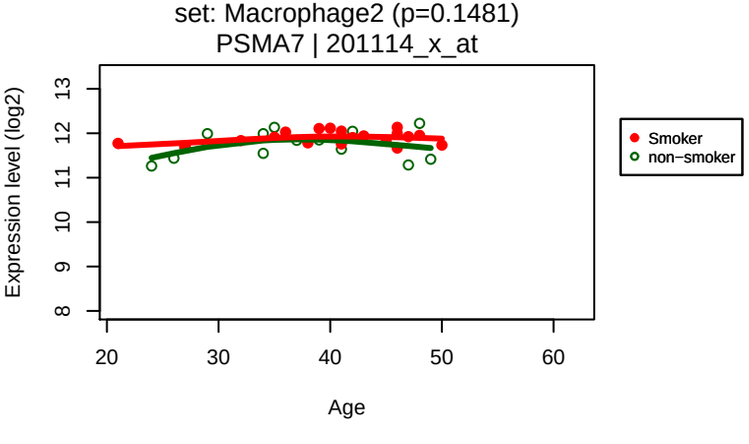

set: SmallAirway2 (p=0.8857)
PSMA7 | 201114_x_at
Expression level (log2)
Age
Smoker
non−smoker

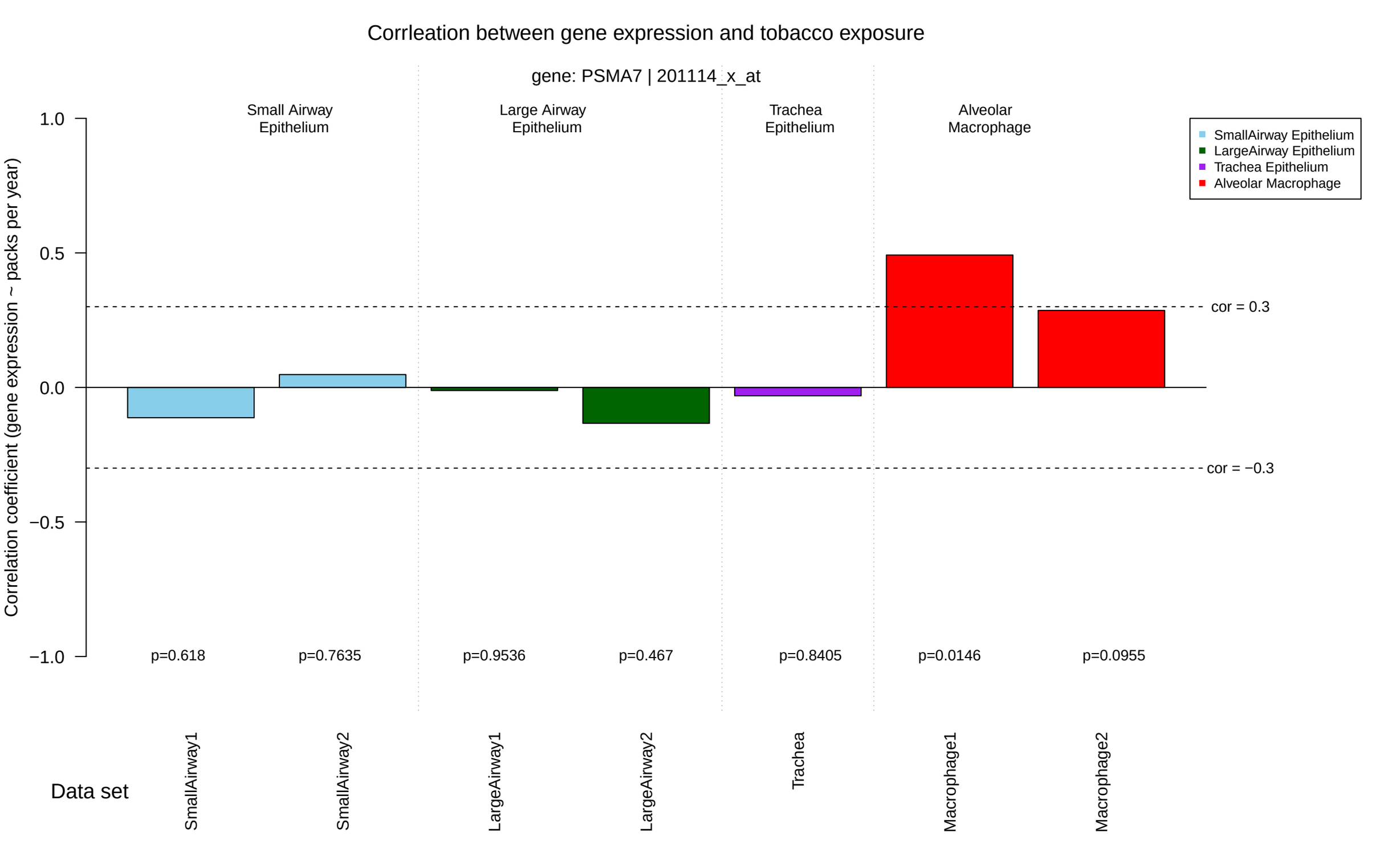

set: LargeAirway1 (p=0.0314)
PSMA7 | 201114_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.561)
PSMA7 | 201114_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7167)
PSMA7 | 201114_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0)
PSMA7 | 201114_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1481)
PSMA7 | 201114_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMA7 | 201114_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.618
p=0.7635
p=0.9536
p=0.467
p=0.8405
p=0.0146
p=0.0955
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2